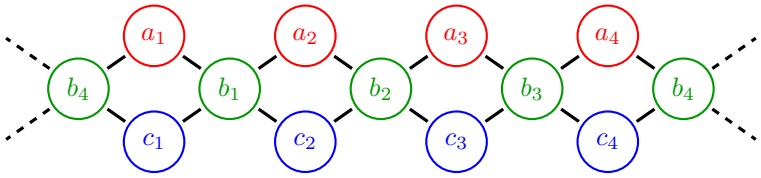
Problem A. Arrange Jewels In Necklace

Time limit: 1 second
Memory limit: 1024 mebibytes

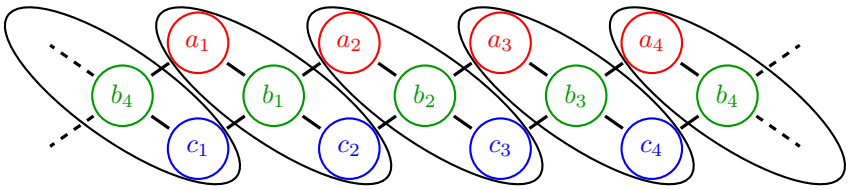
Boris wants to sell beautiful necklaces. He consulted the best designers and jewelers to invent the best necklace. Now he needs your help to understand if there exists a perfect necklace meeting his expectations.

A necklace that Boris wants to make consists of $3k$ jewels, where k is the length of the necklace. Any length $k \geq 2$ is allowed. They are placed in three rows, each having k jewels. All jewels have types (identified with integers). Let us define types of jewels from left to right in the first row as $a_1, a_2, \ldots, a_k$, in the second row as $b_1, b_2, \ldots, b_k$, and in the third row as $c_1, c_2, \ldots, c_k$.

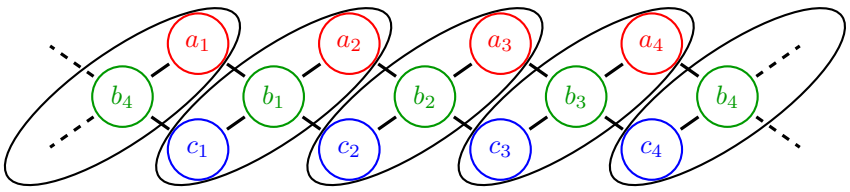


An example of a necklace for $k = 4$

A necklace has k backward columns and k forward columns. Each column contains 3 jewels. The order of jewels in each column is important. Backward columns are $(a_1, b_1, c_2), (a_2, b_2, c_3), \ldots, (a_i, b_i, c_{i+1}), \ldots, (a_k, b_k, c_1)$. Forward columns are $(a_2, b_1, c_1), (a_3, b_2, c_2), \ldots, (a_{i+1}, b_i, c_i), \ldots, (a_1, b_k, c_k)$.



Backward columns



Forward columns

Boris has n options $\{(x_i, y_i, z_i)\}_{i=1}^{n}$ of beautiful backward columns, and m options $\{(u_i, v_i, w_i)\}_{i=1}^{m}$ of beautiful forward columns. He says that a necklace is beautiful if all its columns are beautiful (each column is present in the corresponding beautiful columns set). Find if there exists any beautiful necklace with length ≥ 2.

Also, Boris guarantees that all y_i are different. Note that it is guaranteed only for the middle jewels of beautiful backward columns, so any other numbers may be equal.

Input

Each input consists of multiple test cases. The first line contains a single integer t ($1 \leq t \leq 10^5$) — the number of test cases. Test case descriptions follow.

The first line of each test case contains two integers n, m ($1 \leq n, m \leq 10^5$) — the number of beautiful backward columns and the number of beautiful forward columns.

Each of the next n lines contains three integers x_i, y_i, z_i ($1 \leq x_i, y_i, z_i \leq 10^9$), describing the next beautiful backward column.

Each of the next m lines contains three integers u_i, v_i, w_i ($1 \le u_i, v_i, w_i \le 10^9$), describing the next beautiful forward column.

It is guaranteed that in each test case all y_i are different.

It is guaranteed that the sum of n for all test cases and the sum of m for all test cases does not exceed 10^5.

Output

For each test case, print a single word "YES" if a beautiful necklace with length ≥ 2 exists, and "NO" otherwise.

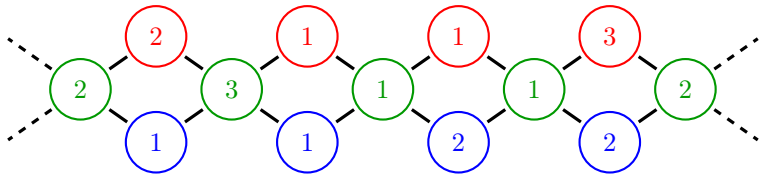
You can print each letter in any case.

Example

standard input	standard output
3 3 5 1 1 2 2 3 1 3 2 1 1 3 1 2 1 2 1 1 1 2 2 2 3 1 2 4 4 2 2 2 1 3 1 2 1 1 4 4 4 1 2 1 1 3 2 1 1 1 2 1 2 1 1 1 1 1 1 1 1	YES NO YES

Note

An example of a beautiful necklace (with length 4) for the first test case:



Problem B. Bean Machine, Reinvented

Time limit:	2 seconds
Memory limit:	1024 mebibytes

Maria has constructed a modified Galton board to show kids some things about probability distributions.

Her board can be represented as a weighted tree rooted at vertex 1. Each time she throws a particle from vertex 1, it follows a random path down the tree to some leaf. Each time it passes through a vertex, it chooses a random edge going down from this vertex, and the probability of choosing an edge is proportional to its weight.

Maria is able to configure the board in any way she wants by changing the weights of the edges. She asked you to process the following queries.

- She changes the weight of some edge.
- She throws a particle from vertex 1 and asks you to calculate the probability of this particle reaching vertex v.

Help Maria with this task.

Input

The first line of input contains one integer n — the number of vertices in the tree ($2 \le n \le 10^5$). The next $n-1$ lines contain the description of the edges of the tree. The i-th of these lines contains the description of the i-th edge in the format "u v w", where u and v are the endpoints of the edge and w is the weight of the edge ($1 \le u, v \le n$, $1 \le w \le 10^4$). It is guaranteed that the edges indeed form a tree.

The next line contains one integer q — the number of queries ($1 \le q \le 2 \cdot 10^5$). Finally, the last q lines of input contain queries, one on each line. Each query is either "1 i w", standing for "change the weight of the i-th edge to w" ($1 \le i \le n-1$, $1 \le w \le 10^4$), or "2 v", meaning "calculate the probability of reaching the vertex v" ($1 \le v \le n$).

Output

For each query of the second type, print one integer, $ab^{-1} \bmod (10^9 + 7)$, where a/b is the irreducible fraction equal to the probability of a particle reaching the vertex v. It can be shown that one can always express this probability in that form.

Examples

standard input	standard output
4 1 2 1 2 3 1 2 4 1 3 2 3 1 2 2 2 3	500000004 666666672
7 1 2 1 1 3 1 2 4 1 2 5 1 3 6 1 3 7 1 3 2 1 2 2 2 4	1 500000004 250000002

Problem C. Captain Thrawn

Time limit: 3 seconds
Memory limit: 1024 mebibytes

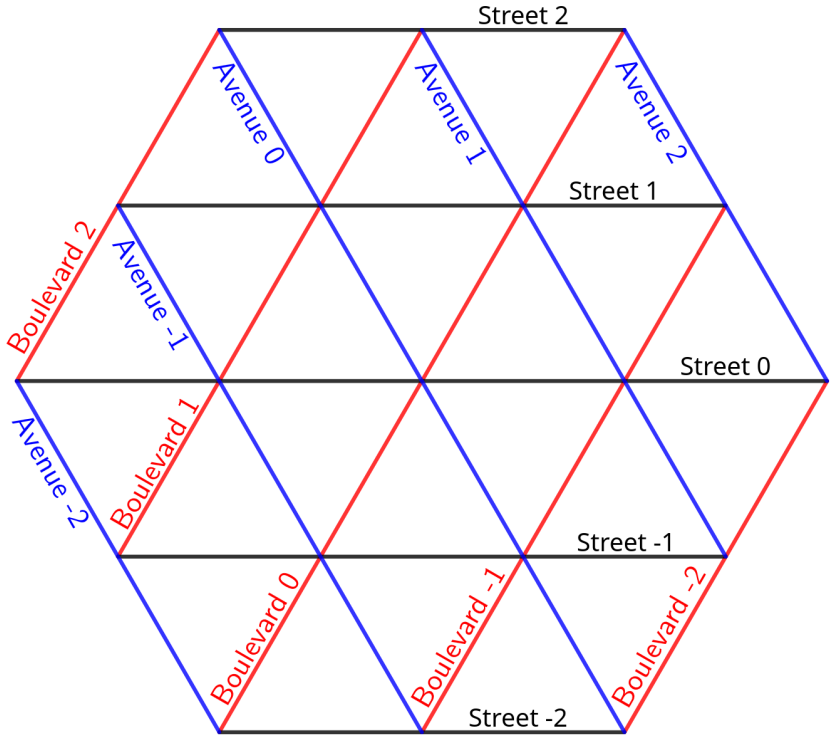
Captain Thrawn of the Chiss Ascendancy is stranded on the Paataatus homeworld! He has a crucial piece of information that he needs to get to Admiral Ar'alani, but he can't wait for a friendly starship to pick him up. Instead, he needs to establish a communications triad to transmit that information to her. The triad consists of 3 parts, which all need to be placed at equal non-zero Euclidean distance from each other (i.e. in the vertices of an equilateral triangle) to successfully transmit data.

However, the political relationship between the Chiss and the Paataatus is tense, so the Paataatus government won't allow Thrawn to put parts of the triad anywhere. The Paataatus capital has a shape of a regular hexagon with the side l. It has $2l + 1$ equally spaced parallel streets going from west to east, numbered from $-l$ to l from south to north, with the street $-l$ going along the southernmost side of the hexagon, the street l going along the northernmost side of the hexagon, and the street 0 going through the center of the hexagon.

Similarly, there are $2l + 1$ equally spaced parallel avenues going from north-west to south-east, numbered from $-l$ to l from south-west to north-east, with the avenue $-l$ going along the south-westernmost side of the hexagon, the avenue l going along the north-easternmost side of the hexagon, and the avenue 0 going through the center of the hexagon.

And there are $2l + 1$ equally spaced parallel boulevards going from south-west to north-east, numbered from $-l$ to l from south-east to north-west, with the boulevard $-l$ going along the south-easternmost side of the hexagon, the boulevard l going along the north-westernmost side of the hexagon, and the boulevard 0 going through the center of the hexagon.

Here is an example of the capital's layout for $l = 2$:



At each intersection of a street, an avenue, and a boulevard, there is a tower. If the street number is x, the avenue number is y, and the boulevard number is z, then the tower has coordinates (x, y, z). Each tower is owned by some Paataatus official.

Luckily for Thrawn, during his career, he has helped several Paataatus officials, and now they are willing

to help him back and allow him to set up a part of his triad in one of their towers. In total, they own n towers, so Thrawn needs to select 3 of them that are equal distance from each other to establish a triad. Before making his choice, Thrawn wants to know all of his options or at least how many there are. Help him count the number of ways to choose 3 towers where a communications triad can be established.

Input

The first line contains two integers l and n — the side length of the capital and the number of towers willing to help Thrawn ($1 \le l \le 500$; $1 \le n \le 3 \cdot l \cdot (l+1) + 1$).

Each of the next n lines contains 3 integers x_i, y_i, z_i — the coordinates of the i-th tower ($-l \le x_i, y_i, z_i \le l$). It is guaranteed that every set of coordinates represents some tower and that all the towers in the input are distinct.

Output

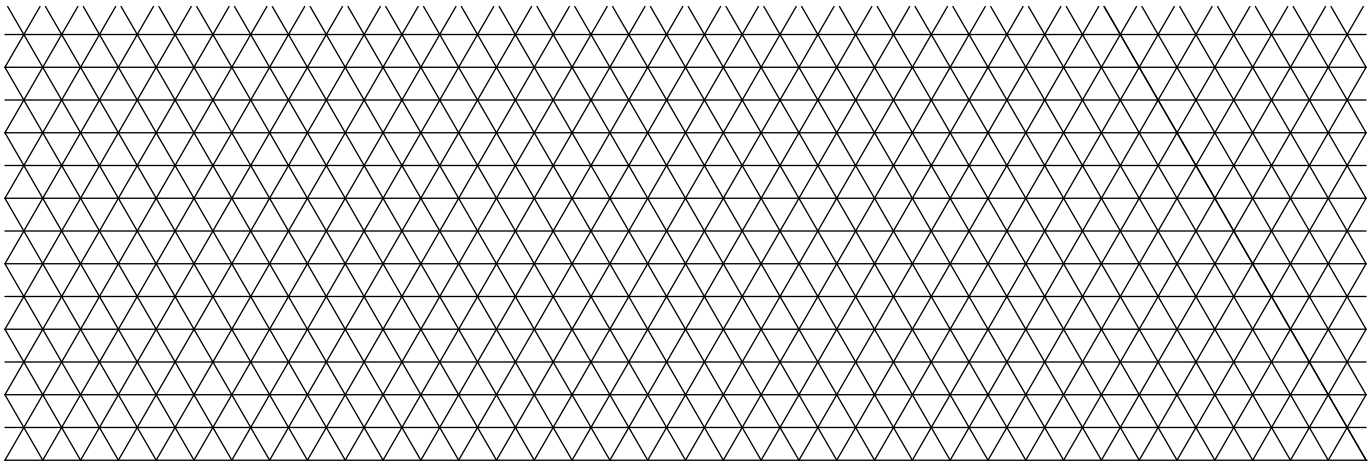
Output a single integer — the number of ways to choose the required 3 towers.

Examples

standard input	standard output
2 6 -1 -2 1 2 2 0 -1 1 -2 -1 0 -1 1 1 0 0 1 -1	1
1 7 0 1 -1 1 1 0 0 -1 1 1 0 1 -1 -1 0 -1 0 -1 0 0 0	8

Note

A triangular grid for your convenience:



Problem D. Divisibility By Factorial

Time limit: 1 second
Memory limit: 1024 mebibytes

In combinatorics, the notation $n!!$ is used to compute the product of all positive integers less than or equal to n that have the same parity. That is, for an odd n, $n!!$ is equal to the product of all odd positive integers not exceeding n, and for an even n — the product of all even positive integers not exceeding n.

Given a number n, find the maximum k such that $n!!$ is divisible by $k!$.

Input

The first line of the input contains a single integer n ($1 \le n \le 10^{18}$).

Output

Output a single integer — the answer to the problem.

Examples

standard input	standard output
1	1
2	2

Problem E. Expected Symmetry

Time limit:	1.5 seconds
Memory limit:	1024 mebibytes

You are given a binary string $s_1 s_2 \ldots s_{2n}$ of length $2n$.

Let's consider following process. Firstly, select an integer k uniformly at random from 1 to n. Now consider prefix $s_1 s_2 \ldots s_{2k}$ of the string s. If it has a symmetry point i ($1 \le i \le k$) such that $s_i = s_{2k-i+1}$, then the process ends immediately. Otherwise, for each i ($1 \le i \le 2k$), the element s_i independently changes its value with probability of $\frac{1}{2}$. After that the process repeats by choosing the next value of k.

Notice that the choice of k is independent at different steps of the process. The events of changing/not changing for bits are also independent between each other.

You should calculate the expected number of steps this process takes to end, including the step at which the process ends.

It can be proven that the answer can be represented as an irreducible fraction $\frac{p}{q}$. You should output $p \cdot q^{-1}$ modulo 998 244 353.

Input

The first line contains a single integer n ($n \in \{10, 10^7\}$).

The second line contains bitwise string of length $2n$: $s_1 s_2 \ldots s_{2n}$.

Output

Output a single integer — the expected number of steps modulo 998 244 353.

Example

standard input	standard output
10 01010000111101010000	222017723

Note

Since the input in this problem is large, we recommend you not to use std::string and std::cin for your C++ implementation.

Problem F. Fast Value Changes

Time limit: 1 second
Memory limit: 1024 mebibytes

Fast Forward has been involved in high-frequency trading for many years. The company's team includes strong developers, Olympiad winners, and past participants of the Moscow Regional Contest. Fast Forward's systems monitor the latest changes in the markets of currencies, cryptocurrencies, precious metals, and various other assets. Therefore, the company's quant researchers actively search for various statistical relationships between various assets: after all, if copper is being actively purchased somewhere, the price of aluminum is likely to rise soon as well.

Recently, quants noticed a pattern between n assets. At time zero, each of these assets has its own price of a_i rubles. The pattern is that the price of asset i approaches the price of asset $(i \bmod n) + 1$ at a rate of 1 ruble per second (on exchanges, the price changes continuously, not every second). When the price of asset i reaches the price of asset $(i \bmod n) + 1$, the price of this asset changes in tandem with the price of asset $(i \bmod n) + 1$ (they always remain equal). If all n assets have the same price, their values will stop changing.

When the prices of all assets are equal, it will be easier for developers to store information about the prices of these assets: they will only need to store a single number. You need to help the Fast Forward developers determine how many seconds it will take for the prices of all n assets to become equal.

Input

The first line contains one integer n — the number of assets ($1 \le n \le 5 \cdot 10^5$).

The second line contains n integers a_1, a_2, ..., a_n — the starting prices of the assets in rubles ($0 \le a_i \le 10^9$).

Output

Output a single real number — the time that will pass before prices equalize. Your output will be considered correct if the absolute or relative error between your output p and the answer q does not exceed 10^{-9}, i.e. $\frac{|p-q|}{\max(q,1)} \le 10^{-9}$.

Examples

standard input	standard output
3 5 8 1	3.5
1 0	0

Note

In the first example, initially, the prices of assets 1 and 3 are rising and the price of asset 2 is falling. At time 1.5, the prices of assets 1 and 2 become equal (with price 6.5) and both begin to fall. The price of asset 3 is 2.5 and still rising. At time 3.5, all prices become equal, and they all stop changing.

In the second example, all the prices are already equal.

Problem G. Gravity Zero Experiments

Time limit: 6 seconds
Memory limit: 1024 mebibytes

Scientists asked for your help to calculate possible outcomes of the experiment with gravity that will be conducted on the International Space Station.

There are n wooden bars fixed in the experiment room. Each bar is a non-empty segment in 3D space. It is guaranteed that no two bars are coplanar — that is, no plane contains both of them.

Initially, there is no gravity in the experiment room. When the experiment starts, gravity will be generated in some gravity direction $\overrightarrow{g}$.

The set of bars $B \subseteq \{1, 2, \ldots, n\}$ is called cyclic if it is possible to place a negligibly small ball on one of the bars at some point, move it with some sequence of operations visiting all bars from set B (any other bars may also be visited, each bar can be visited any number of times), and return to the same initial point.

The ball can be moved using the following operations:

- It can be moved to any other point on the bar it is currently located.
- It can be dropped from the bar; in this case, it will fall down in the gravity direction $\overrightarrow{g}$ and will stop at the nearest point of some other bar. It is allowed to drop the ball only if it will stop.

For every non-empty set of bars $B \subseteq \{1, 2, \ldots, n\}$, calculate if this set can be cyclic for some gravity direction $\overrightarrow{g}$.

Input

Each input consists of multiple test cases. The first line contains a single integer t ($1 \le t \le 1000$) — the number of test cases. Test case descriptions follow.

The first line of each test case contains a single integer n ($1 \le n \le 20$) — the number of bars in the experiment.

Each of the next n lines contains six integers $x_{i1}, y_{i1}, z_{i1}, x_{i2}, y_{i2}, z_{i2}$ ($-10^6 \le x_{i1}, y_{i1}, z_{i1}, x_{i2}, y_{i2}, z_{i2} \le 10^6$), describing the i-th bar. The i-th bar is the segment between points (x_{i1}, y_{i1}, z_{i1}) and (x_{i2}, y_{i2}, z_{i2}). It is guaranteed that $(x_{i1}, y_{i1}, z_{i1}) \neq (x_{i2}, y_{i2}, z_{i2})$.

It is guaranteed that no two bars are coplanar.

Let's define $\max n$ as the maximum value of n among test cases. The following bounds on t are guaranteed depending on $\max n$:

$\max n$	$[1, 6]$	$[7, 10]$	$[11, 15]$	$[16, 20]$
t	≤ 1000	≤ 50	≤ 5	≤ 1

Output

For each test case, print a binary string s of length $2^n - 1$. The value s_i should be 1 if and only if the set B_i corresponding to i is cyclic. The set B_i is the set of positions with 1 in the binary representation of i, i.e. $i = \sum_{b \in B_i} 2^{b-1}$.

Example

standard input	standard output
3 1 -1 1 -1 -1 -1 1 2 0 0 0 1 1 1 0 0 1 1 0 0 3 0 0 0 2 3 3 1 3 0 3 0 3 3 1 0 0 1 3	1 110 1111111

Problem H. Homework Of Xenia

Time limit: 1 second
Memory limit: 1024 mebibytes

Xenia is attending a course on discrete algebraic topological geometry at her university. She skipped a lot of lectures on this topic. To punish Xenia, her teacher gave her the following problem as a homework assignment. The teacher gave Xenia a planar graph G and asked her to embed it into $\mathbb{Z}^{10}$ in such a way that all its vertices are distinct, all coordinates do not exceed 10^6 by absolute value, all edges are segments and have the same length. Note that it is not prohibited for the edges of the embedding to cross.

Help Xenia with this problem.

Input

The first line contains two integers n and m — the number of vertices and edges in the graph ($1 \le n \le 1000$, $0 \le m \le \frac{n(n-1)}{2}$). The next m lines contain the description of the edges. Each of these lines contains two integers u_i and v_i ($1 \le u_i, v_i \le n$) and describes an edge between vertices u_i and v_i.

It is guaranteed that the graph is planar and does not contain loops or multiple edges.

Output

If it is not possible to embed the graph, output a single integer -1.

Otherwise, output n lines — the points in $\mathbb{Z}^{10}$. The i-th of these lines must contain 10 integers $c_{i,1} \ldots c_{i,10}$, $|c_{i,j}| \le 10^6$. These values must satisfy the following property: there exists a positive number d such that for any two vertices u and v connected by an edge in the given graph, $dist(c_u, c_v) = d$. Here, $c_u = (c_{u,1}, \ldots, c_{u,10})$ and $dist(c_u, c_v) = \sqrt{\sum_{i=1}^{10}(c_{u,i} - c_{v,i})^2}$.

Examples

standard input	standard output
2 1 1 2	0 0 0 0 0 0 0 0 0 0 0 5 7 0 1 7 9 0 2 0
4 6 1 2 2 3 2 4 3 4 1 3 1 4	0 0 0 0 0 0 0 0 0 0 1 1 0 0 0 0 0 0 0 0 1 0 1 0 0 0 0 0 0 0 0 1 1 0 0 0 0 0 0 0

Problem I. Interactive Array Modifications

Time limit: 5 seconds
Memory limit: 1024 mebibytes

This is an interactive problem.

The jury has secretly written the array a that contains n integers from 0 to 10^9. It is guaranteed that the minimum value across all elements of a equals 0.

You can make several queries to guess the array. In each query, you should specify an array b that contains n integers from 0 to 10^9. As the response to the query, you receive the new order of elements that should be applied to the array $a + b$ (here $+$ means pointwise sum of arrays a and b) so that it will become stably sorted. In other words, for each query, we can define a new array c of pairs of integers, where $c_i = (a_i + b_i, i)$. The array c is sorted, and you receive the second elements of its pairs in the same order as they appear in the sorted array c.

Your task is to guess the array a using at most 31 queries.

Interaction Protocol

The interaction starts with the jury program writing a single integer t ($1 \le t \le 100$) — the number of scenarios.

The first line of each scenario contains one integer n ($2 \le n \le 10^5$) — the length of the array.

After you read this line, the interaction begins. To ask a query, you should specify an array b of n integers where $0 \le b_i \le 10^9$ and print the line of the following format: "? b_1 b_2 b_3 ... b_n".

As the response, you receive the permutation of integers from 1 to n — the order of elements that should be applied to array a'_i, where $a'_i = a_i + b_i$ for all $1 \le i \le n$, so that a'_i will become stably sorted.

You can ask at most 31 queries of this form for each scenario.

After that, you should print the guessed array a in the following form: "! a_1 a_2 a_3 ... a_n". This line is not considered a query and is not taken into account when counting the number of queries asked.

After this, proceed to the next scenario, or exit if this is the last scenario.

The interactor in this task is not adaptive. In other words, the array a does not change during the interaction.

It is guaranteed that the sum of n among all test cases does not exceed 10^5.

Remember to print the newline and flush the output buffer after each query and after the answer, or your solution will get IL (Idleness Limit Exceeded).

Example

standard input	standard output
2 2 2 1 1 2 2 1 3 1 2 3 2 3 1 1 3 2	 ? 1 0 ? 0 1 ? 1 1 ! 1 0 ? 0 3 1 ? 1 3 1 ? 0 4 1 ! 3 0 2

Problem J. JQU

Time limit: 1 second
Memory limit: 1024 mebibytes

The m-function is a well known function that takes a non-empty string as an argument and cycle shifts each letter of the argument by one. That is, $m(\text{a}) = \text{b}$, $m(\text{z}) = \text{a}$, $m(\text{icpc}) = \text{jdqd}$, etc.

You are given a string consisting of lowercase Latin letters only. You can insert some parentheses into the string and treat some of the letters m as the m-functions. For example, if you are given the string "mama", you can change it to "$m(\text{a}m(\text{a}))$" and treat the letters m as the m-functions to obtain the string "bc". Alternatively, you can change the same string to "$m(\text{am})\text{a}$" and treat only the first letter m as a function to obtain the string "bna". Note that you can't treat m as a function with an empty argument. That is, you can't obtain the string "y" from "my".

What is the lexicographically maximal string that can be obtained from the given string and in how many ways can this maximal string be obtained? As the number of ways can be large, print it modulo 998 244 353.

Input

The only line of the input contains a string s containing only lowercase Latin letters ($1 \le |s| \le 512$).

Output

Print a string and an integer separated by a space — the answer to the problem.

Examples

standard input	standard output
icpc	icpc 1
mma	nb 1
mmyx	nzy 1

Problem K. Know Your Powers

Time limit: 6 seconds
Memory limit: 1024 mebibytes

For a nonnegative integer k and a positive integer n, let $\sigma_k(n)$ denote the sum of k-th powers of all divisors of the number n.

Formally,

$$\sigma_k(n) := \sum_{d \mid n} d^k.$$

You are given n integers $a_1, \ldots, a_n$. Find the value of the following sum:

$$\sum_{1 \le i < j \le n} \sigma_1(a_i + a_j).$$

Input

The first line contains the only integer n ($2 \le n \le 4000$).

The second line contains n positive integers $a_1, \ldots, a_n$ separated by spaces ($1 \le a_i \le 10^9$).

Output

In the only line, print the answer to the problem.

Example

standard input	standard output
5 1 2 3 4 5	91

Problem L. Latin Words

Time limit: 5 seconds
Memory limit: 1024 mebibytes

This is an interactive problem.

A permutation p of letters from the Latin alphabet is secretly written down by the jury, represented as a string of length 26 consisting of distinct uppercase Latin letters.

You can make queries of the following type: for a given substring $p_{i,j}$ from the i-th to the j-th position of the string p inclusive (that is, the string $p_i p_{i+1} \dots p_j$), output the maximum value of a valid Roman numeral that is a substring of $p_{i,j}$.

For example, if the substring is MRC, the output will be 1000 (you can read two numbers: 1000 at positions 1-1, and 100 at positions 3-3; $1000 > 100$). For the substring MIPT, the output will be 1001 (the number at positions 1-2), for the substring HSE — the output will be 0 (no Roman numerals are present), for the substring GCD — the output will be 500 (although there is the string CD at positions 2-3, its value 400 is less than the value of the string D at positions 3-3), and for the substring LCM, the output will be 1000 (valid numbers are present at positions 1-1, 2-2, 3-3, 2-3; the maximum of them is 1000).

Your task is to guess the positions of the letters I, V, X, L, C, D, M, which are used in the representation of Roman numerals, in no more than 19 queries, and to output the lexicographically smallest permutation in which these 7 letters are placed in the required positions.

We remind you of the modern way of writing Roman numerals (the table is taken from Wikipedia):

Place Value	Thousands	Hundreds	Tens	Units
1	M	C	X	I
2	MM	CC	XX	II
3	MMM	CCC	XXX	III
4		CD	XL	IV
5		D	L	V
6		DC	LX	VI
7		DCC	LXX	VII
8		DCCC	LXXX	VIII
9		CM	XC	IX

Note that:

- The numbers 4, 9, 40, 90, 400, and 900 are written in subtractive notation, where the first symbol is subtracted from the second (for example, for 40 (XL), 'X' (10) is subtracted from 'L' (50)). This is the only place where subtractive notation is used.
- A number containing multiple decimal digits is constructed by appending the Roman equivalent of each digit from the most significant to the least significant.
- If the decimal digit is 0, no digits are written in the Roman representation for that place.
- The largest number that can be represented in the Roman numeral system is 3999 (MMMCMXCIX).

Interaction Protocol

The interaction begins with the jury program outputting a single integer t — the number of scenarios ($1 \le t \le 10^4$).

In each scenario, the interaction is started by the participant's program, which outputs a query in the form ? l r, where $0 \le l \le r \le 25$ — the index of the first and last characters in the string p (the characters are numbered from zero). In response, the jury program will output a single integer — the maximum value of a valid Roman numeral that is a substring of $p_{i,j}$. You can make no more than 19 such queries per scenario.

If your program is ready to provide an answer, output !, followed by the string that is the lexicographically smallest permutation of uppercase Latin letters, in which the positions of the letters I, V, X, L, C, D, M correspond to the positions of these letters in the guessed permutation. Outputting the answer does not count as a query. If the answer is output correctly, the interaction proceeds to the next scenario, if there is one, or ends.

It is guaranteed that the original permutation does not change during the interaction, i.e., the interactor is not adaptive.

Example

standard input	standard output
2	? 0 10
1000	? 0 5
500	? 0 6
1000	? 0 1
5	? 0 0
1	? 0 2
10	? 2 3
50	? 4 4
100	! IVXLCDMABEFGHJKNOPQRSTUWYZ ? 10 25
1009	? 10 10
50	? 0 9
0	? 11 11
100	? 11 12
500	? 13 25
1009	? 13 15
1009	? 16 16
5	! ABEFGHJKNOLCDMIXVPQRSTUWYZ

Problem M. Mobile Game Development Lab

Time limit: 1 second
Memory limit: 1024 mebibytes

You are developing a mobile game that uses a very unusual random number generator. This generator receives a positive integer a as input, which depends on the mobile device the game is running on. The generator then writes all the digits in the decimal representations of a and $a + 1$ without leading zeroes into a single string, randomly shuffles them, and uses the resulting string as a sequence of random digits.

You are well aware that MTS Web Services (MWS) is a big tech company providing cloud, AI, and platform solutions for various business needs, and you know many former contest winners who develop products at the company. Now you want to evaluate how often the generator's output string is s, and how the game will perform on the corresponding smartphone. To do this, you need to test your game on a large number of mobile devices. Your game's budget does not allow for this many devices. MWS products help developers in many challenging situations, and this situation is no exception. You've decided to use MWS SunQ — a tool for testing web interfaces and mobile apps for iOS and Android. With this tool, you can test your game on a variety of devices.

Therefore, you're confident that MWS SunQ truly fully covers all possible inputs to the random number generator. But to be sure, you first want to determine how many a exist such that the generator's output on a mobile device will be the string s. Since the number can be very large, output the remainder of its division by 998 244 353.

Input

The first line consists of one integer t ($1 \le t \le 10^4$) — the number of test cases.

Each test case contains on one line and consists of a single string s made up of digits from 0 to 9. The length of each string does not exceed 10^5 characters.

You may assume that the total length of all strings in the input file does not exceed 10^6.

Output

For each test case, output the remainder of the number of integers a from which the given string could have been obtained as a result of applying this operation, modulo 998 244 353.

Example

standard input	standard output
2 2025 121342	0 2